BookLeaf Publishing — Technical Assignment Submission

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GitHub Repository: https://github.com/AnandCh20/BookLeafAssignment
Deliverables: Source Code (Public GitHub Repository), Documentation, and Live Demo

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1. Project Overview & Deliverables

This submission delivers a complete full-stack **Author Support & Communication Portal** for BookLeaf Publishing, featuring:

- **Author Portal (`/author`):** Published titles catalog, royalty earnings/payout tracking (including edge cases like books in typesetting/cover design), support ticket submission with verification attachments, and real-time conversation updates without page refresh.
- **Operations Admin Portal (`/admin`):** Centralized ticket queue with multi-filter sorting, urgent/critical badges, 24h+ overdue flags, AI auto-classification into 6 standard categories, AI priority scoring, Knowledge-Base-grounded AI draft response generation, private internal notes, and team ticket assignment.
- **RESTful API & Interactive Docs (`/api-docs` & `/api/docs`):** Documented REST endpoints with JWT authentication and role-based access control, plus an OpenAPI 3.0 specification.
- **Source Code Repository:** https://github.com/AnandCh20/BookLeafAssignment

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2. Setup Instructions to Run the Project Locally

Prerequisites

- Node.js version 18.0 or higher (v20+ recommended)
- npm (bundled with Node.js)

Step 1: Clone the Repository

```
git clone https://github.com/AnandCh20/BookLeafAssignment.git
cd BookLeafAssignment
```

Step 2: Install Dependencies

```
npm install
```

Step 3: Run the Application

```
npm run dev
```

Open your browser and navigate to:
👉 **http://localhost:3000**

(To run the production build instead: `npm run build && npm start`)

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3. Test Login Credentials & Evaluator Guide

The application comes pre-loaded with the entire sample dataset (10 authors, 18 books, and initial tickets).

Role	Name	Email	Password	Details
Admin	Aakash Mehta	admin@bookleaf.com	admin123	Senior Operations Lead
Admin	Rhea Sen	ops@bookleaf.com	admin123	Author Support Specialist
Author	Priya Sharma	priya.sharma@email.com	password123	2 published titles; ₹3,570 pending payout
Author	Sneha Kulkarni	sneha.kulkarni@email.com	password123	3 titles; 1 in production (*Midnight in Mysore*)
Author	Ananya Reddy	ananya.reddy@email.com	password123	1 title; ₹2,546 pending royalty, ₹0 paid
Author	Kavita Deshmukh	kavita.deshmukh@email.com	password123	2 titles; 1 in typesetting (*Raising Roots*)

Quick Evaluation Feature: The author dashboard includes an **Examiner Switcher** dropdown at the top, allowing you to switch between all 10 authors with a single click without logging out.

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4. Key Notes & Architectural Assumptions

1. Database Architecture & Zero-Friction Evaluation

- **Assumption:** Evaluators should not have to configure external MongoDB Atlas clusters, manage IP whitelists, or resolve connection timeouts to evaluate the assignment.
- **Implementation:** The app includes a built-in document repository pre-seeded with all 10 authors, 18 books, and tickets. A "Reset Sample DB" button is available on the admin console to restore clean test data at any time.
- **Production Readiness:** `mongoose` is installed and configured to connect to `MONGODB_URI` whenever an external cluster is provided.

2. AI Integration & Graceful Degradation

- **Assumption:** In a production publishing system processing 1,200+ monthly books, external LLM APIs may experience network timeouts or rate limits (HTTP 429). The system must never crash or block ticket creation.
- **Dual-Layer Architecture:**
  1. *Primary:* Calls OpenAI `gpt-4o-mini` if `OPENAI_API_KEY` is provided in environment variables.
  2. *Graceful Fallback:* If no API key is supplied, an intelligent domain knowledge base engine automatically classifies the ticket into the 6 standard categories, scores priority, and generates an empathetic draft grounded in BookLeaf policies (80/20 royalty split, quarterly 45-day cycle, ₹1,000 threshold, Delhi warehouse 5-7 day print turnaround, 24-48h stock sync).
- **Cost Consciousness:** Instead of dumping full document histories into prompts, the system performs selective context injection based on inquiry intent to minimize token usage.

3. Real-Time Communication on Serverless (Vercel)

- **Trade-off:** Stateful WebSockets often experience dropouts on serverless platforms like Vercel unless routed through external persistent socket servers.
- **Solution:** A 3.5-second polling loop paired with optimistic UI updates ensures author screens update in near real-time when an admin responds, without requiring manual page refreshes.

4. Code Standards & Separation of Concerns

- **Zero Comments:** All code strictly omits inline and block comments per project guidelines.
- **Server Component Boundary:** App Router page routes (`src/app/**/page.js`) are pure Server Components; interactive UI logic is cleanly encapsulated in Client Components under `src/components/`.
- **Monochrome Theme:** Implemented in Vanilla CSS with high-contrast black and white styling, responsive across mobile, tablet, and desktop viewports.

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5. Technical Assignment 2: Royalty Automation & No-Code System

In addition to the custom full-stack application (Assignment 1), the repository includes the complete deliverables for **Assignment 2 (Author Royalty Dashboard & Automated Notification System)**:

Deliverables Manifest

1. **n8n Automation Workflow Export:**
   - [`assignment-2/n8n_royalty_summary_workflow.json`] (file:///Users/karishma_sharma/Development/Assignment/assignment-2/n8n_royalty_summary_workflow.json) (Production 9-node workflow with webhook triggers, data aggregation, GPT prompt, error handling, and callbacks).
2. **Relational Database Seed Datasets (CSV):**
   - [`assignment-2/data/authors.csv`] (file:///Users/karishma_sharma/Development/Assignment/assignment-2/data/authors.csv) (10 normalized author records ready for 1-click Bubble.io import).
   - [`assignment-2/data/books.csv`] (file:///Users/karishma_sharma/Development/Assignment/assignment-2/data/books.csv) (18

book catalog records with foreign keys and financial metrics).

3. **Automated Test Suite & Output Samples:**
   - [assignment-2/test_n8n_simulation.py] (file:///Users/karishma_sharma/Development/Assignment/assignment-2/test_n8n_simulation.py) (Automated test runner verified against all 10 authors).
   - [assignment-2/simulated_output_samples.json] (file:///Users/karishma_sharma/Development/Assignment/assignment-2/simulated_output_samples.json) (Pre-generated sample email statements).
4. **Comprehensive Architecture Blueprint & PDF Report:**
   - Markdown: [ASSIGNMENT-2-DOCUMENTATION.md] (file:///Users/karishma_sharma/Development/Assignment/ASSIGNMENT-2-DOCUMENTATION.md)
   - Formal PDF Deliverable: [ASSIGNMENT-2-DOCUMENTATION.pdf] (file:///Users/karishma_sharma/Development/Assignment/ASSIGNMENT-2-DOCUMENTATION.pdf) (Includes Bubble schema, privacy rules, status badge conditionals, and security specs).